Daisuke Yamada

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Education

University of Wisconsin–Madison — Madison, WI
PhD in Computer Science — Expected Graduation June 2028
Advised by Prof. Vikas Singh

Carleton College — Northfield, MN
Bachelor of Arts: Mathematics (Honors) and Computer Science (Honors) — June 2023
GPA: 3.95/4.00

DIS Study Abroad in Scandinavia — Copenhagen, Denmark
Semester Abroad in Machine Learning — January to May 2023

Budapest Semesters in Mathematics — Budapest, Hungary
Semester Abroad in Mathematics and Computer Science — August to December 2021

Publications

- Learning Goal-Reaching Quasimetric Geometry From Finite-Time Reachability
  Daisuke Yamada, Travis Pence, Vikas Singh
  Preprint
- FORGE: Procedural Weight Generation for Sparse MoE Inference
  Chanyoung Moon, Travis Pence, **Daisuke Yamada**, Jiaqi Mo, Vikas Singh, Karthikeyan Sankaralingam
  ASPLOS 2027
- Fast Sandwich Products in Clifford Algebra
  Travis Pence, **Daisuke Yamada**, Vikas Singh
  NeurIPS 2026
- Adaptive Scoring and Thresholding with Human Feedback for Robust Out-of-Distribution Detection
  Daisuke Yamada, Harit Vishwakarma, Ramya Korlakai Vinayak
  TMLR 2026 · NeurIPS 2026 (J2C Track)
- Tree-Structured Orthonormal Decomposition of the Aitchison Simplex
  Daisuke Yamada, Qijun Zhang, Travis Pence, Barbara Bendlin, Federico Rey, Vikas Singh
  ICML 2026
- Recursive Binding on a Budget: Subspace Carving in Order-p Tensor Memories
  Travis Pence, **Daisuke Yamada**, Vikas Singh
  ICML 2026
- Composing Linear Layers from Irreducibles
  Travis Pence, **Daisuke Yamada**, Vikas Singh
  NeurIPS 2025
- Selecting Balls From Urns With Partial Replacement Rules
  Julian Burden, Chandramauli Chakraborty, Qizhou Fang, Lisa Lin, Nasser Malibari, Sammi Matoush, Isaiah Milbank, Zahan Parekh, Martín Prado, Rachael Ren, Qizhao Rong, Maximiliano Sánchez Garza, Eli Sun, Enrique Treviño, **Daisuke Yamada**
  PUMP Journal, 2024

Experience

University of Wisconsin–Madison — Madison, WI
PhD Research Assistant — Fall 2023 – Present

Carleton College — Northfield, MN
Undergraduate Research Assistant — Fall 2022

MIAC Analytics — New York City, NY
Machine Learning Engineer Intern — Winter 2021, Summer 2022

Polymath Jr. REU — Remote
REU Research — Summer 2021

Service & Teaching

Reviewer
NeurIPS (Top Reviewer, 2026), ICLR, Annals of Applied Statistics — 2025 – Present

Teaching Assistant — Madison, WI
UW–Madison — 2023 – Present

- Data Science Programming in Python (CS 220)
- Discrete Mathematics (CS 240)
- Data Programming I (CS 320)
- Numerical Linear Algebra (CS 513)

Scholarships and Awards

- Thesis Distinction in Computer Science at Carleton College — 2023
- Thesis Distinction in Mathematics at Carleton College — 2023
- Patricia V. Damon Scholarship — 2022
- Dean's Lists at Carleton College — 2022, 2023